CO1-AT1	Name : Bastina Sri S. Reg. no. : 1921260231 Course : Machine Learning. Course Code : ITA0611

A Factory has three boxes containing coloured balls. Box A contains 6 red, 6 blue, and 2 green balls; Box B contains 5 red, 7 blue and 3 green balls; and Box C contains 8 red, 2 blue and 5 green balls. One box is selected at random, but B box is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Given Box B is twice as likely as A and C :

$$P(A) = 1/4 , \quad P(B) = 1/2 , \quad P(C) = 1/4$$

Probabilities of drawing a blue ball :

$$P(\text{Blue}/A) = \frac{4}{12} = \frac{1}{3}$$

$$P(\text{Blue}/B) = 7/15.$$

$$P(\text{Blue}/C) = 2/15.$$

using Bayes' theorem,

$$P(C/Blue) = \frac{P(C)\,P(Blue/C)}{P(A).P(Blue/A) + P(B).P(Blue/B) + P(C).P(Blue/}$$

$$= \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{7}{15} + \frac{1}{4} \cdot \frac{2}{15}}$$

$$= \frac{2}{21}$$

$$\approx 9.52\%$$

A parking facility has three sections containing cars of different color. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other section due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

$P(A) = 1/4$; $P(B) = 1/2$, $P(C) = 1/4$.

$$P(\text{Blue}/A) = 3/10.$$

$$P(\text{Blue}/B) = \frac{6}{15} = \frac{2}{5}.$$

$$P(\text{Blue}/C) = 2/15.$$

$$P(C/\text{Blue}) = \frac{\frac{1}{4} \cdot \frac{2}{15}}{\frac{1}{4} \cdot 3/10 + 1/2 \cdot 2/5 + 1/4 \cdot 2/15}$$

$$= \frac{1/30}{3/40 + \frac{2}{5} + \frac{1}{30}} = \frac{4}{37}.$$

$\therefore 10.81\%$.

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbours algorithm is applied for binary classification. The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210 True Positives, 190 Negative and 30 False Positives. Based on the given information, determine the missing False negative values, and then calculate the accuracy, Precision, Sensitivity, and Spicificity of the classifier, explaining its effectiveness in medical diagnosis.

$$1800 + 1800 = 3600.$$

20% is used for testing

$$3600(0.2) = 720.$$

Given:

$TP = 210$. $TN = 190$ $FP = 30$ $FN = ?$

Since the total test samples are 720:

$$TP + TN + FP + FN = 720.$$

$$210 + 190 + 30 + FN = 720.$$

$$FN = 290.$$

4. Examine the performance of a machine learning model using its training and validation loss over multiple epochs:

a) Epoch 1:

Training loss = 2.8, Validation loss = 2.4

Epoch 2:

Training loss = 2.5, Validation loss = 2.2.

Epoch 3:

Training loss = 2.2, Validation loss = 2.0.

Epoch 4:

Training loss = 2.0, Validation loss = 2.1

Epoch 5:

Training loss = 1.8, Validation loss = 2.3

Epoch 6:

Training loss = 1.6, Validation loss = 2.6

Epoch 7:

Training loss = 1.5, Validation loss = 2.9

Epoch 8:

Training loss = 1.3, Validatieis loss = 3.2.

i) Identify the epoch at which mode begins to overfit.

ii) Suggest. Explain why validation despite this.

Accuracy = $\frac{TP + TN}{TP + TN + FP + FN}$

$= \frac{210 + 190}{720}$

$= 55.56\%$

Precision = $\frac{TP}{TP + FN}$

$= \frac{210}{210 + 290}$

$= 42\%$ $= 87.50\%$

Sensitivity = $\frac{TP}{TP + FN}$

$= \frac{210}{210 + 290}$

$= 42.00\%$

Specificity = $\frac{TN}{TN + FP}$

$= \frac{190}{190 + 30}$

$= 86.36\%$

ii) ~~Suggest~~.

Explain why validation accuracy decreases despite training accuracy increasing.

The model is memorizing the training data instead of learning patterns that generalize to unseen data. Therefore, its performance on the training data improves, but its performance on validation data gets worse. This is called overfitting.

iii) Suggest two methods to improve validation performance.

1. Use regularization such as dropout or L2 Eg regularization to reduce overfitting.

2. Use early stopping by stopping training when validation accuracy stops improving.

5. Examine the performance of a machine learning model using training and validation accuracy over multiple epochs.

a) Epoch 1: Training Accuracy = 60%, Validation Accuracy = 58%.

Epoch 2:
Training Accuracy = 65%, Validation Accuracy = 63%.

Epoch 3:
Training Accuracy = 75%, , Validation Accuracy = 68%.

Epoch 4:
Training Accuracy = ~~80~~ 75%, , Validation Accuracy = 72%.

Epoch 5:
Training Accuracy = 80%., Validation Accuracy = 73%.

Epoch 6:
Training Accuracy = 85%. , Validation Accuracy = 72%.

Epoch 7:
Training Accuracy = 88%. , Validation Accuracy = 70%.

Epoch 8:
Training Accuracy = 92%., Validation Accuracy = 68%

i) Identify the epoch at which the model starts overfitting.

The model starts overfitting at Epoch 5.

It keeps decreasing

2.8 → 2.5 → 2.2 → 2.0 → 1.8 → 1.6 → 1.5 → 1.3.

It decreases until Epoch 3.

2.4 → 2.2 → 2.0.

Then it starts increasing:

2.0 → 2.1 → 2.3 → 2.6 → 2.9 → 3.2.

∴ Overfitting begins at Epoch 4.

Suggest two effective techniques to reduce overfitting in this model.

1. Early stopping - Stop training when validation loss starts increasing.
2. Regularization - Use techniques such as L1/L2 Regularization or dropout.

Effect on generalization

Overfitting means the model learns the training data too closely, including its noise and specific patterns. As a result, it performs well on training data but performs poorly on unseen / test data.